Going beyond ODE systems with the moment-SOS hierarchy

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Topics

Modular framework: only need to swap out dynamics

Examples in this talk:

- Discrete-time systems
- Hybrid systems
- Stochastic systems

Not PDEs.

Previously on Didier's Talk...

Optimal Control Intro

Basic scheme:

$$V^*(\cdot) = \inf_u \int_{t_0}^{T} [\text{Running cost}(t)]dt + \text{Terminal Cost}$$

Dynamics are obeyed

u is a valid control

State constraints (initial, running, final)

Template for future OCP

Optimal Control Problems (ODE)

Filling in the mathematics (value function V^*)

$$
\begin{aligned}
V^*(t_0, x_0) = \inf_u \int_{t_0}^{T} & J(t, x(t), u(t))dt + J_T(x(T)) \\
& \dot{x} = f(t, x(t), u(t)) && \forall t \in [t_0, T] \\
& u(t) \in \mathcal{U} && \forall t \in [t_0, T] \\
& x(t_0) = x_0,\ x(T) \in X_T
\end{aligned}
$$

Optimal control (if it exists) is $u^*(t)$

Dynamical Systems Problems

Other common tasks in dynamical systems:

- Reachable set estimation
- Peak estimation
- Global attractor description
- Invariant measure discovery
- Distance estimation
- Region of attraction maximization

Can plug your dynamics in and get answers

Where do dynamics appear? (function side)

Hamilton-Jacobi Bellman conditions in $V^*(t,x)$

$$V^*(T,x) = J_T(x) \text{ in } X_T \tag{1a}$$
$$\inf_u \, (\partial_t + f \cdot \nabla_x) V(t,x,u) + J(t,x,u) = 0 \tag{1b}$$

Subvalue $v \leq V^*$, function LP ($\sup v(0,x_0)$)

$$v(T,x) \leq J_T(x) \text{ in } X_T \tag{2a}$$
$$(\partial_t + f \cdot \nabla_x) v(t,x,u) + J(t,x,u) \geq 0 \tag{2b}$$

Moment-SOS: restrict $v \in \mathbb{R}[t,x]$, do SOS tightening

Where do dynamics appear? (measure side)

Liouville equation holds for all $v \in C^1([0, T] \times X)$:

$$\text{[Value at] End} = \text{Start} + \text{Accumulated Change} \tag{3}$$

$$v(T, x(T \mid x_0)) = v(0, x_0) + \int_{t'=0}^{t} (\partial_t + f \cdot \nabla_x) v(t', x(t' \mid x_0)) dt'$$

Expressed in measures for (μ_0, μ_T, μ)

$$\begin{aligned} \langle v(T, x), \mu_T(x) \rangle = \langle v(0, x), \mu_0(x) \rangle \\ + \langle (\partial_t + f \cdot \nabla_x) v(t, x), \mu(t, x, u) \rangle \end{aligned} \tag{4}$$

Moment-SOS: affine constraints on pseudo-moments

Other dynamics?

Basic take-aways

Swap out the term $(\partial_t + f \cdot \nabla_x)$ for your system's **generator**

The mathematical expressions look the same...

but the underlying theorems are different

Discrete-Time Dynamics

Clock instances $t \in 0, 1, 2, \ldots$

$$x_{t+1} = R(t, x_t, u_t) \tag{5}$$

Control actions are usually discrete time:

- HVAC: building actuation at 15 minute intervals
- Power dispatch: day-ahead planning
- Robotics: Computation time (clock frequency, sensors)

Discrete-Time Optimal Control

Almost the same mathematics

$$V^*(t_0, x_0) = \inf_u \sum_{t_0}^{T-1} J(t, x(t), u(t)) + J_T(x(T))$$

$$x(t+1) = R(t, x(t), u(t)) \qquad \forall t \in [t_0, T]$$

$$u(t) \in \mathcal{U} \qquad \forall t \in [t_0, T]$$

$$x(t_0) = x_0, \;\; x(T) \in X_T$$

Optimal control (if it exists) is $u^*(t)$

Generator (Deterministic)

Generator: Incremental change in the test function v

Continuous-Time system

$$\dot{x} = f(t, x, u) \qquad v \mapsto (\partial_t + f \cdot \nabla_x)v \tag{6}$$

Discrete-Time

$$x_+ = R(t, x, u) \qquad v \mapsto v(t+1, R(t, x, u)) - v(t, x) \tag{7}$$

Subvalue comparison

Just swap out generators [1]

In continuous-time:

$$0 \leq (\partial_t + f \cdot \nabla_x) v(t, x, u) + J(t, x, u) \tag{8}$$

In discrete-time:

$$0 \leq v(t+1, R(t, x, u)) - v(t, x) + J(t, x, u) \tag{9}$$

[1] Han, W., & Tedrake, R. (2018, October). Controller synthesis for discrete-time polynomial systems via occupation measures. In 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (pp. 6911-6918). IEEE.

Computational Complexity

Consider $f, R \in \mathbb{R}[t, x, u]$

Degree bounds given test function $v \in \mathbb{R}[t, x]$:

$$\deg(\partial_t + f \cdot \nabla_x)v = \deg v + \deg f - 1$$
$$\deg(v(t+1, R(t,x,u)) - v(t,x)) = \deg v * \deg R$$

Adding a new lifting variable $\tilde{x} = R(t, x, u)$ [2]

$$\deg(v(t+1, \tilde{x}) - v(t,x)) = \deg v$$

Discrete-time: much larger degree, or doubled state dimension

[2] Miller, Jared, and Mario Sznaier. "Analysis and Control of Input-Affine Dynamical Systems using Infinite-Dimensional Robust Counterparts." arXiv:2112.14838 (2021).

Further Considerations

Properties based on continuity, convexity, boundedness

1. No relaxation gap to OCP [3] [4]
2. Uniqueness of solution (usually fails when controlled) [5]
3. Strong duality

[3] Vinter, R. B., & Lewis, R. M. (1978). The equivalence of strong and weak formulations for certain problems in optimal control. SIAM Journal on Control and Optimization, 16(4), 546-570.

[4] Hernández-Lerma, O., & Lasserre, J. B. (2012). Discrete-time Markov control processes: basic optimality criteria (Vol. 30). Springer Science & Business Media.

[5] Henrion, D., & Korda, M. (2013). Convex computation of the region of attraction of polynomial control systems. IEEE Transactions on Automatic Control, 59(2), 297-312.

Hybrid Systems

Hybrid Systems Introduction

Continuous-time dynamics with discrete transitions [6]

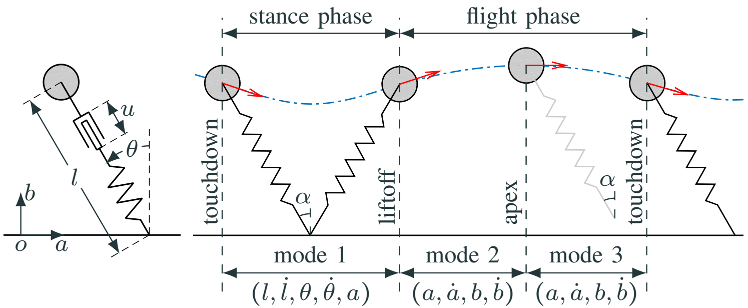
Examples:

1. Bouncing ball
2. Walking motion
3. Gearbox in car

Hazard of Zeno behavior

[6] R. Goebel, R. G. Sanfelice, and A. R. Teel, "Hybrid Dynamical Systems," IEEE Control Systems Magazine, vol. 29, no. 2, pp. 28–93, 2009

Example hybrid system



Spring-Loaded Inverted Pendulum (SLIP)[7]

[7] Zhao, P., Mohan, S., & Vasudevan, R. (2019). Optimal control of polynomial hybrid systems via convex relaxations. IEEE Transactions on Automatic Control, 65(5), 2062-2077. (Figure 2 in the paper)

Hybrid Dynamics

Modes $\ell \in 1..L$ with arcs $e = (\ell, \ell')$

Continuous dynamics f_ℓ

$$\dot{x}_\ell(t) = f_\ell(t, x(t), u(t)) \tag{10a}$$

Discrete resets R_e at *guard surfaces* S_e (e.g. ground)

$$x_{\mathrm{dst}(e)}(t^+) = R_e(t, x_{\mathrm{src}(e)}(t), u_e(t)) \tag{10b}$$

Introduce subvalue v_ℓ at each location

Hybrid Optimal Control (function)

Supremize $v_{\mathrm{Loc}(t=0)}(0, x(0))$ subject to:

Continuous-time running costs J_ℓ

$$(\partial_t + f_\ell \cdot \nabla_x) v_\ell(t, x_\ell, u) + J_\ell(t, x_\ell, u) \geq 0$$

Terminal costs at time T

$$J_{T\ell}(x_\ell) \geq v_\ell(T, x_\ell)$$

Reset costs on jumps G_e

$$G_e(t, x_{\mathrm{src}(e)}, u_e) \geq v_{\mathrm{src}(e)}(t, x_{\mathrm{src}(e)}) - v_{\mathrm{dst}(e)}(t, R_e(t, x_{\mathrm{src(e)}, u_e}))$$

Hybrid Liouville Equation

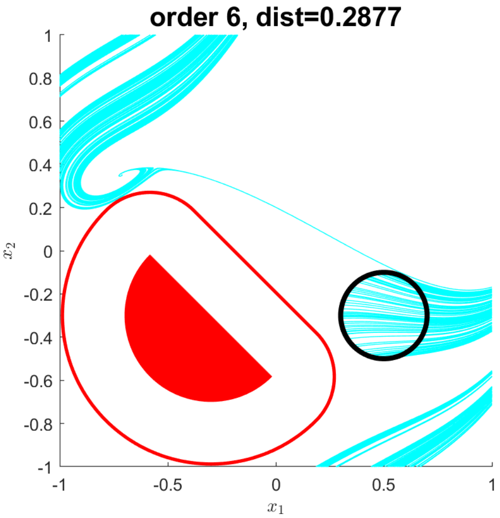
Per-location measures $(\mu_{0\ell}, \mu_{T\ell}, \mu_\ell)$, guard measures ρ_e

[Mass in mode ℓ] - [flow from ℓ] + [flow to ℓ] $= 0$

For all test functions $v_\ell \in C^1$:

$$\begin{aligned}\langle v(T,x), \mu_{T\ell}(x)\rangle =& \langle v(0,x), \mu_{0\ell}(x)\rangle + \langle (\partial_t + f_\ell \cdot \nabla_x) v_\ell(t, x_\ell, u), \mu_\ell\rangle \\ & - \sum_{\text{source}(e)=\ell} \langle v_\ell(t, x_\ell), \rho_e(t, x_\ell, u_e)\rangle \\ & + \sum_{\text{destination}(e)=\ell} \langle v_\ell(t, R(t, x_\ell, u_e)), \rho_e(t, x_\ell, u_e)\rangle\end{aligned}$$

Hybrid Example

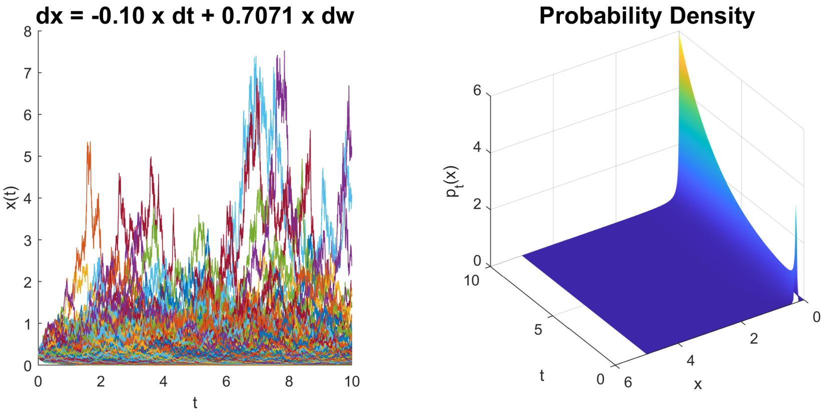


Distance estimation given boundary jumps

Stochastic Systems

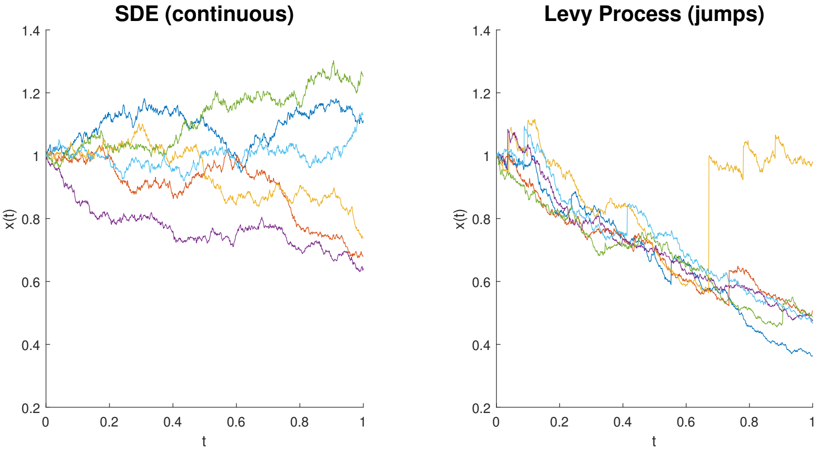
Stochastic Process

A collection of time-indexed probability distributions $\{\mu_t\}$



SDE: $dx = f(t,x)dt + g(t,x)dw$ (Itô)

Stochastic Process Examples



Geometric Brownian motion (left), Merton jump diffusion (right)

Stochastic Optimal Control Problems (ODE)

Insert expectations in cost

$$
\begin{aligned}
P^*(x_0) = \inf_u \int_{t=0}^{T} \mathbb{E}[J(t, x(t), u(t))]dt + \mathbb{E}[J_T(x(T))] \\
& x(t) \text{ follows Stoch. Proc.} && \forall t \in [0, T] \\
& u(t) \in \mathcal{U} && \forall t \in [0, T] \\
& x(0) \sim \mu_0 \\
& \text{Chance constraints on } \; X, \; X_T
\end{aligned}
$$

Risk-sensitive: replace $\mathbb{E}$ with CVaR, mean-variance

The Process for Stochastic Processes

Describe stochastic process by a **generator** $\mathcal{L}$

Increment-τ change ($\forall v \in \mathrm{dom}(\mathcal{L})$):

$$\mathcal{L}_\tau v = \lim_{\tau' \to \tau^+} \frac{\mathbb{E}[v(t+\tau', x) \mid \mu_{t+\tau'}] - v(t, x)}{\tau'}$$

(we restrict only to processes with generators)

Martingale Relation

Conservation law for trajectories $\{\mu_t\}_{t=0}^{T}$ (with $x(t) \sim \mu_t$)

Discrete-Time ($\tau = 1$)

$$\mathbb{E}_{x\sim\mu_T}[v(T,x)] = \mathbb{E}_{x\sim\mu_0}[v(0,x)] + \sum_{t=0}^{T-1} \mathbb{E}_{x\sim\mu_t}[\mathcal{L}_1 v(t,x,u)]$$

Continuous-Time

$$\mathbb{E}_{x\sim\mu_T}[v(T,x)] = \mathbb{E}_{x\sim\mu_0}[v(0,x)] + \int_{t=0}^{T} \mathbb{E}_{x\sim\mu_t}[\mathcal{L}_0 v(t,x,u)]dt$$

Occupation Measures (if invariant)

$$\langle v(T,x), \mu_T(x)\rangle = \langle v(0,x), \mu_0(x)\rangle + \langle \mathcal{L} v(t,x), \mu(t,x,u)\rangle$$

Examples of Generators

Discrete-time Markov Process

$$X_{t+1} = F(t, X_t, u_t, \omega_t), \qquad \omega_t \sim \xi \text{ (sampled)}$$
$$\mathcal{L}_1 v = \left(\int_\Omega v\,(t+1, F(t, x, u, \omega))\, d\xi(\omega) - v \right)$$

Example: ξ is white noise

Examples of Generators (cont.)

Stochastic Differential Equation

$$dx = f(t, x, u)dt + g(t, x, u)dW,$$
$$\mathcal{L}_0 v = \partial_t v + f \cdot \nabla_x v + g^T(\nabla^2_{xx} v)g/2$$

related to Kolmogorov/Fokker-Planck forward equation

Can add jumps: Lévy process

Stochastic formulation

Given generator $\mathcal{L}$ of stochastic process[8]

Liouville $\rightarrow$ Martingale by using $\mathcal{L}$ (measure)

$\langle v(T,x), \mu_T(x)\rangle = \langle v(0,x), \mu_0(x)\rangle + \langle \mathcal{L}v(t,x,u), \mu(t,x,u)\rangle$

Subvalue decrease condition (function)

$$0 \leq \mathcal{L}v(t,x,u) + J(t,x,u)$$

[8] O. Hernández-Lerma, J.B. Lasserre. Discrete-Time Markov Control Processes: Basic Optimality Criteria. Springer Verlag, New York, 1996.

Stochastic formulation (finishing touches)

Stochastic expectation performed in objective:

$$\int_{t=0}^{T} \mathbb{E}[J(t, x(t), u(t))]dt + \mathbb{E}[J_T(x(T))]$$
$$\to \langle J(t, x, u), \mu(t, x, u)\rangle + \langle J_T(x), \mu_T(x)\rangle$$

Moment-SOS: $\mathcal{L}$ maps $\mathbb{R}[t, x] \to \mathbb{R}[t, x, u]$

(risk-sensitive control usually harder)

Stochastic Hybrid Systems

Each location has a generator $\mathcal{L}_\ell$

Stochastic reset maps R_ℓ

All measure/function constraints identical.

Take-aways

Conclusion

Linear programming framework is modular

Can substitute in your preferred dynamics...

...so long as you have a **generator** (and it's not a PDE)

Then truncate and solve!

Thanks!